Year 10 Foundation

REVISION PACK

Graphs 2

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 2 · Conversion graphs (Lesson 8.1)

Q	Working	Answer
using a conversion graph		
1.1	$30\text{ KD} = €90$ (from the graph) $€90 \times 10 = €900$	**(a) €60** **(b) 30 KD** **(c) €900**
1.2	$40\text{ KD} = £100$ (from the graph) $£100 \times 10 = £1000$	**(a) £40** **(b) 28 KD** **(c) £1000**
1.3	$20\text{ KD} = \text{AED } 240$ (from the graph) $\text{AED } 240 \times 10 = \text{AED } 2400$	**(a) AED 180** **(b) 8 KD** **(c) AED 2400**
1.4	$15\text{ KD} = \text{SAR } 60$ (from the graph) $\text{SAR } 60 \times 10 = \text{SAR } 600$	**(a) SAR 100** **(b) 15 KD** **(c) SAR 600**

Graphs 2 · Distance-time graphs (Lesson 8.2)

Q	Working	Answer
reading a distance-time graph		
2.1	$35 - 20 = 15$ $8\text{ km in }20\text{ min}$ $8 \div \frac{1}{3} = 24$	**(a) 15 minutes** **(b) 4 km** **(c) 24 km/h**
2.2	$60 - 40 = 20$ $10\text{ km in }30\text{ min}$ $10 \div \frac{1}{2} = 20$	**(a) 20 minutes** **(b) 5 km** **(c) 20 km/h**
2.3	$45 - 15 = 30$ $6\text{ km in }15\text{ min}$ $6 \div \frac{1}{4} = 24$	**(a) 30 minutes** **(b) 2 km** **(c) 24 km/h**
2.4	$50 - 30 = 20$ $9\text{ km in }30\text{ min}$ $9 \div \frac{1}{2} = 18$	**(a) 20 minutes** **(b) 3 km** **(c) 18 km/h**

Graphs 2 · Speed-time graphs (Lesson 8.3)

Q	Working	Answer
acceleration and constant speed		
3.1	$(a)\ 10 \div 5 = 2$ $(b)\ (15 - 5) \times 10 = 100$	**(a) 2 m/s²** **(b) 100 m**
3.2	$(a)\ 12 \div 4 = 3$ $(b)\ (10 - 4) \times 12 = 72$	**(a) 3 m/s²** **(b) 72 m**
3.3	$(a)\ 18 \div 6 = 3$ $(b)\ (16 - 6) \times 18 = 180$	**(a) 3 m/s²** **(b) 180 m**
3.4	$(a)\ 20 \div 10 = 2$ $(b)\ (20 - 10) \times 20 = 200$	**(a) 2 m/s²** **(b) 200 m**

Graphs 2 · Mid-points and perpendicular lines (Lessons 8.4, 8.5)

Q	Working	Answer
mid-point of a line		
4.1	$\left(\frac{4+10}{2}, \frac{1+7}{2}\right)$	**(7, 4)**
4.2	$\left(\frac{1+9}{2}, \frac{6+12}{2}\right)$	**(5, 9)**
4.3	$\left(\frac{3+9}{2}, \frac{2+12}{2}\right)$	**(6, 7)**
4.4	$\left(\frac{2+6}{2}, \frac{4+10}{2}\right)$	**(4, 7)**
4.5	$\left(\frac{0+8}{2}, \frac{3+5}{2}\right)$	**(4, 4)**
4.6	$\left(\frac{5+11}{2}, \frac{1+9}{2}\right)$	**(8, 5)**
perpendicular lines		

4.7	**(a)** $-1 \div 2 = -\frac{1}{2}$ **(b)** $m_1 = 4,\ m_2 = -\frac{1}{4}$ $4 \times (-\frac{1}{4}) = -1$	**(a)** $-\frac{1}{2}$ **(b) gradients multiply to −1, so perpendicular**
4.8	**(a)** $-1 \div 6 = -\frac{1}{6}$ **(b)** $m_1 = 5,\ m_2 = -\frac{1}{5}$ $5 \times (-\frac{1}{5}) = -1$	**(a)** $-\frac{1}{6}$ **(b) gradients multiply to −1, so perpendicular**
4.9	**(a)** $-1 \div \frac{1}{3} = -3$ **(b)** $m_1 = \frac{1}{2},\ m_2 = -2$ $\frac{1}{2} \times (-2) = -1$	**(a)** -3 **(b) gradients multiply to −1, so perpendicular**
4.10	**(a)** $-1 \div 3 = -\frac{1}{3}$ **(b)** $m_1 = 6,\ m_2 = -\frac{1}{6}$ $6 \times (-\frac{1}{6}) = -1$	**(a)** $-\frac{1}{3}$ **(b) gradients multiply to −1, so perpendicular**
4.11	**(a)** $-1 \div \frac{1}{4} = -4$ **(b)** $m_1 = -5,\ m_2 = \frac{1}{5}$ $-5 \times \frac{1}{5} = -1$	**(a)** -4 **(b) gradients multiply to −1, so perpendicular**
4.12	**(a)** $-1 \div (-2) = \frac{1}{2}$ **(b)** $m_1 = \frac{2}{3},\ m_2 = -\frac{3}{2}$ $\frac{2}{3} \times (-\frac{3}{2}) = -1$	**(a)** $\frac{1}{2}$ **(b) gradients multiply to −1, so perpendicular**